

CSE 715: Computer Graphics

Chapter 4 — 2D Transformations

Complete Model Answers (2020–2024)

University of Chittagong — 7th Semester B.Sc. (Engg.) Examination

Source: Schaum's Outline of Computer Graphics (Xiang & Plastock), Chapter 4

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Theoretical Framework — Reusable Across All Years

This cluster shows up as **Q3 (2020–2023)** or **Q4 (2020 Group-A / 2024)** every year — the exact question number shifts (2020 splits it oddly: the numeric rotation/scaling pair is its Q4, while 8-way symmetry is bundled into the same Q4 as a Ch3 sub-part). The content is always: derive/apply a 2D transformation matrix (rotation and/or scaling), usually about an *arbitrary fixed point* rather than the origin, plus a recurring theory question distinguishing coordinate systems or transformation types.

The Four Basic 2D Transformations (Homogeneous Matrix Form)

Using homogeneous coordinates $(x, y, 1)$ so translation becomes a matrix multiply like the others: $[x' \ y' \ 1] = [x \ y \ 1] \cdot T$.

Transformation	Matrix T (row-vector convention, post-multiply)
Translation by (t_x, t_y)	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$
Rotation by θ (CCW, about origin)	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Scaling by (s_x, s_y) (about origin)	$\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Reflection about x -axis	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Reflection about y -axis	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Reflection about origin (=180° rotation)	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
x -shear (factor sh_x)	$x' = x + sh_x \cdot y, \quad y' = y$
y -shear (factor sh_y)	$x' = x, \quad y' = y + sh_y \cdot x$

Equivalently, without homogeneous coordinates: $x' = x \cos \theta - y \sin \theta$, $y' = x \sin \theta + y \cos \theta$ (rotation); $x' = s_x \cdot x$, $y' = s_y \cdot y$ (scaling about origin).

Deriving the Rotation Matrix from Geometry

Let $P(x, y)$ be at distance l from the origin, at angle α to the x -axis, so $x = l \cos \alpha$, $y = l \sin \alpha$. Rotating P by θ (CCW) about the origin gives $P'(x', y')$ at the same distance l , angle $\alpha + \theta$:

$$x' = l \cos(\alpha + \theta) = l \cos \alpha \cos \theta - l \sin \alpha \sin \theta = x \cos \theta - y \sin \theta$$

$$y' = l \sin(\alpha + \theta) = l \sin \alpha \cos \theta + l \cos \alpha \sin \theta = x \sin \theta + y \cos \theta$$

using the angle-sum identities and substituting back $l \cos \alpha = x$, $l \sin \alpha = y$.

Transformations About an Arbitrary Fixed Point (the recurring exam pattern)

3-step recipe (works identically for rotation *and* scaling about a fixed point $F(f_x, f_y)$):

1. **Translate** F to the origin: subtract F from every point, $(x - f_x, y - f_y)$.
2. **Apply** the transformation about the origin (rotate by θ , or scale by (s_x, s_y)) to the translated coordinates.
3. **Translate back**: add F to the result.

Composite matrix: $T(F) \cdot R(\theta) \cdot T(-F)$ for rotation, or $T(F) \cdot S(s_x, s_y) \cdot T(-F)$ for scaling. Closed forms used throughout this chapter:

Rotate about F : $x' = f_x + (x - f_x) \cos \theta - (y - f_y) \sin \theta$, $y' = f_y + (x - f_x) \sin \theta + (y - f_y) \cos \theta$

Scale about F : $x' = f_x + s(x - f_x)$, $y' = f_y + s(y - f_y)$

Geometric Transformation vs. Coordinate Transformation

Geometric transformation	Coordinate transformation
The object itself moves/changes (position, size, orientation) while the reference axes stay fixed.	The object stays physically fixed ; instead the reference frame (axes) is moved/rotated/scaled, and the object's coordinates are re-expressed relative to the new frame.
Produces a genuinely new object (new points in the same coordinate system).	Produces the same object described in a different coordinate system — no physical change.
Uses matrix M directly on point coordinates: $P' = M \cdot P$.	Mathematically equivalent to applying M^{-1} to describe the same fixed point relative to the transformed frame.

Why a Device-Independent Coordinate System is Needed

Physical displays vary in resolution and physical size across systems. If a picture were defined directly in device (pixel) coordinates, it would look different — dis-

torted, cropped, or mis-scaled — on every different device. The fix is a **Normalized (Device-Independent) Coordinate System**: define/model the picture once in a standard normalized range (e.g. the unit square $[0, 1] \times [0, 1]$), then apply a separate **workstation/viewport transformation** that maps normalized coordinates to whatever the actual device's pixel range happens to be. This decouples *what* is drawn from *which device* it is drawn on, so the same picture definition renders correctly (same relative proportions) everywhere.

Examination 2024 — Q4(a–e) (Section A)

Question breakdown:

- (a) Given a shape in a 2D coordinate system, Figure (a), mention the transformations used on (a) to generate the shapes in (b), (c), (d), and (e). (2)
- (b) A point $P(x, y)$ is rotated at angle θ about the origin O in CCW direction. Derive the rotation matrix using the given figure. (2)
- (c) Translate the line segment P_1P_2 by -1 units in x and -2 units in y ; $P_1(1, 2)$, $P_2(3, 3)$. Find the new endpoint coordinates. (2)
- (d) Mention the steps needed to rotate an arbitrary line about a point P_1 . (2)
- (e) Write down the matrices for shear transformation in 2D space. (1)

(a) — Identify the Transformations from the Figure [2 marks]

The figure shows shape (a) (a small face icon near the axis origin) reproduced as four variants (b)–(e) at different positions/orientations. **Method (apply this to your own paper's exact figure)**: compare each variant's position and orientation against (a) using these four signatures:

- ▷ **Pure translation**: same size, same orientation (face right-side-up, not mirrored), only the position has moved — e.g. variant (e), which sits diagonally opposite (a) but is not flipped.
- ▷ **Reflection about the x -axis**: the shape is flipped **upside-down** (vertically mirrored) but not left-right reversed — appears directly below/above (a) on the opposite side of the horizontal axis.
- ▷ **Reflection about the y -axis**: the shape is flipped **left-right** (horizontally mirrored) but not upside-down — appears on the opposite side of the vertical axis at the same height.
- ▷ **Reflection about the origin (=180° rotation)**: the shape is flipped **both** vertically and horizontally — appears diagonally opposite through the origin.

Reading this paper's layout (four quadrant positions (b)/(c)/(d) along the bottom row and (e) at top-right, mirroring (a) at top-left/top-center): (b) and (d) are the two single-axis reflections (x -axis and y -axis respectively, whichever one is upside-down

vs. left-right flipped in your printed copy), (c) is reflection about the origin (both flips combined, sitting at bottom-middle), and (e) is a pure translation (same orientation as (a), just moved). **On exam day: name each transform by checking orientation, not position** — position alone is ambiguous, but "is it flipped vertically / horizontally / both / neither" uniquely identifies translation vs. the three reflection types.

(b) — Derive the Rotation Matrix [2 marks]

See §1.2 for the full derivation. Final result in matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

(c) — Translate $P_1(1, 2), P_2(3, 3)$ by $(-1, -2)$ [2 marks]

$$P'_1 = (1 - 1, 2 - 2) = (0, 0) \quad P'_2 = (3 - 1, 3 - 2) = (2, 1)$$

(d) — Steps to Rotate an Arbitrary Line About a Point P_1 [2 marks]

Same recipe as §1.3, applied to a line (both endpoints get the same treatment):

1. Translate the whole line so that P_1 coincides with the origin: subtract P_1 's coordinates from both endpoints.
2. Rotate the translated endpoints by the required angle θ about the origin using the standard rotation matrix.
3. Translate the result back by adding P_1 's original coordinates to both rotated endpoints.

Composite matrix: $T(P_1) \cdot R(\theta) \cdot T(-P_1)$.

(e) — 2D Shear Transformation Matrices [1 mark]

$$x\text{-shear: } \begin{bmatrix} 1 & 0 & 0 \\ sh_x & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} (x' = x + sh_x \cdot y, y' = y) \quad y\text{-shear: } \begin{bmatrix} 1 & sh_y & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} (x' = x, y' = y + sh_y \cdot x)$$

Examination 2023 — Q3(a–c) (Section A)

Question breakdown:

- (a) While capturing/displaying images, monitor sizes vary between systems. What kind of coordinate system is required to resolve this, and why? (1)
- (b) Find the new coordinates of triangle $P(-1, 2)$, $Q(4, 4)$, $R(1, -4)$ after 45 rotation for: (i) about the origin, (ii) about the point $Q(4, 4)$. (4)
- (c) Given a 3D object with points $A(3, 0, 3)$, $B(3, 3, 6)$, $C(3, 0, 1)$, $D(0, 0, 0)$, apply scaling parameters 3 (X), 2 (Y), 3 (Z) and find the new coordinates. (4)

(a) — Coordinate System Rationale [1 mark]

See §1.5. A **normalized (device-independent) coordinate system** decouples the image definition from any one device's resolution/size.

(b) — 45° Rotation, About Origin and About $Q(4, 4)$ [4 marks]

$\cos 45 = \sin 45 \approx 0.70711$.

(i) About the origin — $x' = x \cos 45 - y \sin 45$, $y' = x \sin 45 + y \cos 45$:

Point	$x - y$	$x + y$	$P' = (0.70711(x - y), 0.70711(x + y))$
$P(-1, 2)$	-3	1	$(-2.1213, 0.7071)$
$Q(4, 4)$	0	8	$(0, 5.6569)$
$R(1, -4)$	5	-3	$(3.5355, -2.1213)$

(ii) About $Q(4, 4)$ — translate by $-Q$, rotate 45, translate back by $+Q$:

Point	$(x - 4, y - 4)$	rotate 45	$+Q(4, 4)$
$P(-1, 2)$	$(-5, -2)$	$(-2.1213, -4.9497)$	$(1.8787, -0.9497)$
$Q(4, 4)$	$(0, 0)$	$(0, 0)$	$(4, 4)$ (<i>fixed point — unchanged</i>)
$R(1, -4)$	$(-3, -8)$	$(3.5355, -7.7782)$	$(7.5355, -3.7782)$

(c) — 3D Scaling [4 marks] (*Related — primary home is Ch6/Tier-5 "3D scaling application," answered here too since it is bundled in this year's Q3*)

New coordinates = $(3x, 2y, 3z)$ (scaling about the origin, since no other fixed point is specified and $D = (0, 0, 0)$ already sits at the origin):

$$A(3, 0, 3) \rightarrow (9, 0, 9) \quad B(3, 3, 6) \rightarrow (9, 6, 18) \quad C(3, 0, 1) \rightarrow (9, 0, 3) \quad D(0, 0, 0) \rightarrow (0, 0, 0)$$

Examination 2022 — Q3(a–c) (Section A)

Question breakdown:

- (a) While capturing/displaying images, monitor sizes vary. What kind of coordinate system is required and why? (1)
- (b) Find the new coordinates of triangle $P(-1, 2), Q(2, 4), R(0, 0)$ keeping $Q(2, 4)$ fixed for: (i) expanded thrice its size, (ii) reduced to half its size. (4)
- (c) Find the matrix to perform a 45 rotation of triangle $A(5, 6), B(2, 1), C(5, 3)$: (i) about the origin, (ii) about the point $(2, 3)$. (4)

(a) — Coordinate System Rationale [1 mark]

Identical question and answer to 2023(a) — see §1.5.

(b) — Scale Triangle Keeping $Q(2, 4)$ Fixed, $\times 3$ and $\times 0.5$ [4 marks]

Formula (§1.3): $P' = Q + s(P - Q)$.

Point	(i) $s = 3: Q + 3(P - Q)$	(ii) $s = 0.5: Q + 0.5(P - Q)$
$P(-1, 2)$	$(2, 4) + 3(-3, -2) = (-7, -2)$	$(2, 4) + 0.5(-3, -2) = (0.5, 3)$
$Q(2, 4)$	fixed: $(2, 4)$	fixed: $(2, 4)$
$R(0, 0)$	$(2, 4) + 3(-2, -4) = (-4, -8)$	$(2, 4) + 0.5(-2, -4) = (1, 2)$

(c) — 45° Rotation, About Origin and About $(2, 3)$ [4 marks]

About the origin:

Point	$x - y$	$x + y$	$A' = (0.70711(x - y), 0.70711(x + y))$
$A(5, 6)$	-1	11	$(-0.7071, 7.7782)$
$B(2, 1)$	1	3	$(0.7071, 2.1213)$
$C(5, 3)$	2	8	$(1.4142, 5.6569)$

About $(2, 3)$ — translate by $-(2, 3)$, rotate 45, translate back:

Point	$(x - 2, y - 3)$	rotate 45	$+(2, 3)$
$A(5, 6)$	$(3, 3)$	$(0, 4.2426)$	$(2, 7.2426)$
$B(2, 1)$	$(0, -2)$	$(1.4142, -1.4142)$	$(3.4142, 1.5858)$
$C(5, 3)$	$(3, 0)$	$(2.1213, 2.1213)$	$(4.1213, 5.1213)$

Examination 2021 — Q3(a–c) (Section A)

Question breakdown:

- (a) Differentiate between the geometric and coordinate transformation. (1.25)
- (b) Magnify the Pentagon ARTHE to twice its size while keeping $E(1, 6)$ fixed. $A(-4, 3), R(-2, -5), T(2, -3), H(7, 0), E(1, 6)$. (3.5)
- (c) Perform a 60 rotation of the rectangle $A(1, 1), B(1, 2), C(2, 2), D(2, 1)$ about the point $E(-1, -1)$. (4)

(a) — Geometric vs. Coordinate Transformation [1.25 marks]

See §1.4 for the full comparison table.

(b) — Magnify Pentagon ARTHE $\times 2$ Keeping $E(1, 6)$ Fixed [3.5 marks]

Formula: $P' = E + 2(P - E)$.

Point	$(x - 1, y - 6)$	$\times 2$	$+E(1, 6)$
$A(-4, 3)$	$(-5, -3)$	$(-10, -6)$	$(-9, 0)$
$R(-2, -5)$	$(-3, -11)$	$(-6, -22)$	$(-5, -16)$
$T(2, -3)$	$(1, -9)$	$(2, -18)$	$(3, -12)$
$H(7, 0)$	$(6, -6)$	$(12, -12)$	$(13, -6)$
$E(1, 6)$	—	—	fixed: $(1, 6)$

(c) — 60 Rotation About $E(-1, -1)$ [4 marks]

$\cos 60 = 0.5$, $\sin 60 = 0.86603$. Formula (§1.3): $x' = f_x + (x - f_x) \cos \theta - (y - f_y) \sin \theta$, $y' = f_y + (x - f_x) \sin \theta + (y - f_y) \cos \theta$.

Point	$(x - (-1), y - (-1))$	rotate 60 (relative)	$+E(-1, -1)$
$A(1, 1)$	$(2, 2)$	$(-0.7321, 2.7321)$	$(-1.7321, 1.7321)$
$B(1, 2)$	$(2, 3)$	$(-1.5981, 3.2321)$	$(-2.5981, 2.2321)$
$C(2, 2)$	$(3, 3)$	$(-1.0981, 4.0981)$	$(-2.0981, 3.0981)$
$D(2, 1)$	$(3, 2)$	$(-0.2321, 3.5981)$	$(-1.2321, 2.5981)$

Examination 2020 — Q4(a–c), plus related Q5(d) & Q6(a) (Group-A / Group-B)

Question breakdown:

- Q4(a) Perform a 90 rotation of rectangle $A(1, 1), B(1, 2), C(2, 2), D(2, 1)$ about the point $B(-1, -1)$. (4.25)
- Q4(b) Magnify the triangle $A(1, 1), B(2, 1), C(2, 2)$ to twice its size while keeping $B(2, 1)$ fixed. (3.5)
- Q4(c) Show 8-way symmetry of a circle. (1) — *Ch3 topic, already solved in Chapter3_Solutions.pdf; cross-referenced only.*
- Q5(d) Differentiate between the geometric and coordinate transformation. (part of Q5, Group-B)
- Q6(a) While capturing/displaying images, monitor sizes vary. What kind of coordinate system is required and why? (part of Q6, Group-B)

Q4(a) — 90 Rotation About $B(-1, -1)$ [4.25 marks]

Note: this fixed point is named "B" in the question text but is a different point from the rectangle's own vertex $B(1, 2)$ — treat it as an independently-given pivot, exactly like E in the 2021 paper. $\cos 90 = 0, \sin 90 = 1 \Rightarrow x' = f_x - (y - f_y), y' = f_y + (x - f_x)$.

Point	$(x - (-1), y - (-1))$	rotate 90 (relative)	$+(-1, -1)$
$A(1, 1)$	$(2, 2)$	$(-2, 2)$	$(-3, 1)$
$B(1, 2)$	$(2, 3)$	$(-3, 2)$	$(-4, 1)$
$C(2, 2)$	$(3, 3)$	$(-3, 3)$	$(-4, 2)$
$D(2, 1)$	$(3, 2)$	$(-2, 3)$	$(-3, 2)$

Q4(b) — Magnify Triangle $\times 2$ Keeping $B(2, 1)$ Fixed [3.5 marks]

Formula: $P' = B + 2(P - B)$.

$$A(1, 1) \rightarrow (2, 1) + 2(-1, 0) = (0, 1) \quad B(2, 1) \rightarrow \text{fixed: } (2, 1) \quad C(2, 2) \rightarrow (2, 1) + 2(0, 1) = (2, 3)$$

Q4(c) — 8-Way Symmetry [1 mark]

See Chapter3_Solutions.pdf §1.2/§3 — one octant point (x, y) mirrors to 7 others.

Q5(d) — Geometric vs. Coordinate Transformation [related]

Same answer as 2021(a) — see §1.4.

Q6(a) — Coordinate System Rationale [related]

Same answer as 2022(a)/2023(a) — see §1.5.

Quick Summary — Exam Patterns for Chapter 4

Pattern	Years	Method to memorise
Rotate points/polygon by θ , about origin <i>and</i> about an arbitrary fixed point	2020, 2021, 2022, 2023, 2024	Origin: $x' = x \cos \theta - y \sin \theta$, $y' = x \sin \theta + y \cos \theta$. Arbitrary point F : subtract F , rotate, add F back
Magnify/scale a polygon keeping one named vertex fixed	2020, 2021, 2022, 2023	$P' = F + s(P - F)$ for every vertex; the fixed vertex itself is unchanged
Coordinate-system rationale (why device-independent coords)	2020, 2022, 2023	One-line answer: normalized coords decouple the picture from any one device's resolution; a separate workstation transform maps to the real device
Geometric vs. Coordinate transformation (definitional)	2020, 2021	Object moves (geometric) vs. axes move (coordinate) — same matrix, opposite thing is "held still"
Figure-based transform ID + translation + shear + composite-rotation steps (2024 "extras")	2024 only	Read orientation cues (flipped how?) for reflections; translation is a direct coordinate subtraction; shear multiplies one axis by the other's value
3D scaling (bundled into this Q3/Q4 slot in 2023)	2023	Same per-axis scale rule extended to z : $(s_x x, s_y y, s_z z)$

Exam-day strategy for the 2D Transformations cluster: every numeric sub-part reduces to the same 3-step "translate–transform–translate back" recipe (§1.3) — memorise that recipe once and it covers rotation, scaling, and (per 2024) rotating an arbitrary line about a point. The 1-mark theory sub-parts (coordinate-system rationale, geometric vs. coordinate transformation) are free points if the one-paragraph distinctions in §1.4/§1.5 are memorised — don't skip them chasing only the numeric marks.